

Analytic Number Theory

Lecture Notes

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Preface

These are ongoing lecture notes for the undergraduate topics course in introductory analytic number theory taught at Seoul National University in Fall 2026. The material draws heavily from M. Ram Murty's *Problem in Analytic Number Theory* and Tom M. Apostol's *Introduction to Analytic Number Theory*.

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1 Arithmetic functions

Definition 1.1.

1. $\mathbb{N} = \{1, 2, 3, \dots\}$, $\mathbb{P} = \{p \in \mathbb{N} : p \text{ prime}\}$
2. For $x \in \mathbb{R}$, write $[x] = \max\{n \in \mathbb{Z} : n \leq x\}$ and $\{x\} = x - [x] \in [0, 1)$.

Definition 1.2. Let $X \subseteq \mathbb{R}$. Let $f, g : X \rightarrow \mathbb{C}$ be functions.

1. Assume $g \geq 0$ is real nonnegative. We write

$$f(x) = O(g(x)) \quad \text{or} \quad f(x) \ll g(x)$$

if there is a constant $K > 0$ such that $|f(x)| \leq Kg(x)$ for all $x \in X$.

2. For $a \in \mathbb{R} \cup \{\pm\infty\}$, we write:

- $f(x) = o(g(x))$ as $x \rightarrow a$ if $\lim_{x \rightarrow a} f(x)/g(x) = 0$.
- $f(x) \sim g(x)$ as $x \rightarrow a$ if $\lim_{x \rightarrow a} f(x)/g(x) = 1$.

Definition 1.3. An *arithmetic function* is a function $f : \mathbb{N} \rightarrow \mathbb{C}$.

Example 1.4.

1. Constant functions, identity function $\text{id}_{\mathbb{N}}(n) = n$.
2. Let $S \subseteq \mathbb{N}$. The characteristic function of S is given by

$$\delta_S(n) = \begin{cases} 1 & n \in S \\ 0 & \text{otherwise.} \end{cases}$$

3. Divisor function $d(n) = \#\{d \in \mathbb{N} : d \mid n\} = \sum_{d \mid n} 1$.
4. $(k \in \mathbb{N})$ $d_k(n) = \#\{(d_1, \dots, d_k) \in \mathbb{N}^k : d_1 \cdots d_k = n\}$. ($d_1 = 1$, $d_2 = d$.)
5. $(s \in \mathbb{C})$ $\sigma_s(n) = \sum_{d \mid n} d^s$. (Note. $\sigma_0 = d$.)
6. $(p \in \mathbb{P})$ $v_p(n) = \max\{k \in \mathbb{Z}_{\geq 0} : p^k \mid n\}$ p -adic valuation.
7. $\omega(n) = \#\{p \in \mathbb{P} : p \mid n\}$
8. $\Omega(n) = \sum_p v_p(n) = \#$ of prime divisors counted with multiplicity.
9. Liouville function: $\lambda(n) = (-1)^{\Omega(n)}$.

2 Summatory functions

An important theme in analytic number theory is to study the behavior of an arithmetic function on average.

Definition 2.1. The *summatory function* of $a = (a_n)_{n \in \mathbb{N}}$ is $S_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{C}$,

$$S_a(x) = \sum_{n \leq x} a_n.$$

Example 2.2.

1. $\sum_{n \leq x} 1 = [x] = x - \{x\} = x + O(1)$.
2. Suppose $\lim_{n \rightarrow \infty} a_n = \alpha \in \mathbb{C}$. Then $S_a(x) = (\alpha + o(1))x$.
3. Let $a_n = (-1)^n$. Then $S_a(x) = o(x)$.

Example 2.3. Define

$$\pi(x) = \#\{p \leq x : p \text{ prime}\} = S_{\delta_{\mathbb{P}}}(x).$$

Then prime number theorem, conjectured by Legendre and Gauss in early 1800's and proved in 1896 independently by Hadamard and de La Vallée Poussin, states that

$$\pi(x) \sim \frac{x}{\log x} \sim \int_2^x \frac{dt}{\log t} \quad \text{as } x \rightarrow \infty.$$

Definition 2.4. The Möbius function is given by

$$\mu(n) = \begin{cases} (-1)^{\omega(n)} & n \text{ squarefree (i.e., } m^2 \mid n \implies m = 1) \\ 0 & \text{otherwise.} \end{cases}$$

PNT is equivalent to $M(x) = \sum_{n \leq x} \mu(n) = o(x)$. The celebrated Riemann hypothesis is equivalent to $M(x) = O_{\epsilon}(x^{1/2+\epsilon})$.

3 Partial summation

Exercise 3.1. Let $a, b : \mathbb{N} \rightarrow \mathbb{C}$ be arithmetic functions. If $N \in \mathbb{N}$, then

$$S_{ab}(N) = S_a(N)b_N - \sum_{n < N} S_a(n)(b_{n+1} - b_n).$$

Proof. Let $T_{a,b}(N)$ be the right hand side. Clearly $S_{ab}(1) = T_{a,b}(1)$. It suffices to show that $S_{ab}(N) - S_{ab}(N-1) = T_{a,b}(N) - T_{a,b}(N-1)$ for $N \geq 2$. LHS is $a_N b_N$, while

$$\begin{aligned} T_{a,b}(N) - T_{a,b}(N-1) &= S_a(N)b_N - S_a(N-1)b_{N-1} \\ &\quad - S_a(N-1)(b_N - b_{N-1}) \\ &= (S_a(N) - S_a(N-1))b_N = a_N b_N. \end{aligned} \quad \square$$

Theorem 3.2. Let $a : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function. If $[y, x] \subseteq \mathbb{R}_{\geq 0}$ and $f \in C^1([y, x])$,

$$\sum_{y < n \leq x} a_n f(n) = [S_a(x)f(x)]_y^x - \int_y^x S_a(t)f'(t) dt.$$

Proof. Both sides are linear in $a_1, \dots, a_{[x]}$. We may thus assume $a_n = \delta_k(n)$ for some $k \leq x$.

1. ($k \leq y$) $0 = f(x) - f(y) - \int_y^x f'(t) dt$
2. ($y < k \leq x$) $f(k) = f(x) - \int_k^x f'(t) dt.$

□

Example 3.3.

$$\sum_{n \leq x} \log n = x \log x - x + O(\log x).$$

Proof. Let $a_n \equiv 1$ and $f(x) = \log x$. We have $S_a(x) = [x] = x + O(1)$ so

$$\begin{aligned} \sum_{n \leq x} \log n &= (x + O(1)) \log x - \int_1^x (t + O(1)) \frac{1}{t} dt \\ &= x \log x + O(\log x) - \int_1^x dt + O\left(\int_1^x \frac{1}{t} dt\right) \\ &= x \log x - x + O(\log x). \end{aligned} \quad \square$$

4 Euler's formula

Corollary 4.1 (Euler's formula). *If $[y, x] \subseteq \mathbb{R}_{\geq 0}$ and $f \in C^1([y, x])$,*

$$\sum_{y < n \leq x} f(n) = \int_y^x f(t) dt - [\{t\}f(t)]_y^x + \int_y^x \{t\}f'(t) dt.$$

Proof. Applying partial summation with $a_n \equiv 1$,

$$\begin{aligned} \sum_{y < n \leq x} f(n) &= [[t]f(t)]_y^x - \int_y^x [t]f'(t) dt \\ &= \left([tf(t)]_y^x - \int_y^x tf'(t) dt \right) - [\{t\}f(t)]_y^x + \int_y^x \{t\}f'(t) dt. \end{aligned}$$

The parenthetical expression is $\int_y^x f(t) dt$ by integration by parts. $\square$

Example 4.2. There is a constant $0 < \gamma < 1$ (called Euler's constant) such that

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + O\left(\frac{1}{x}\right).$$

Proof.

$$\begin{aligned} \sum_{n \leq x} \frac{1}{n} &= 1 + [\log t]_1^x - \left[\frac{\{t\}}{t} \right]_1^x - \int_1^x \frac{\{t\}}{t^2} dt \\ &= \log x + 1 - \int_1^\infty \frac{\{t\}}{t^2} dt - \int_x^1 \frac{\{t\}}{t^2} dt + O\left(\frac{1}{x}\right). \end{aligned}$$

The result follows by setting $\gamma = 1 - \int_1^\infty \frac{\{t\}}{t^2} dt$. $\square$

Exercise 4.3. Let $N(R) = \{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 \leq R^2\}$. Prove that

$$N(R) = \pi R^2 + O(R).$$

Conjecture 4.4 (Gauss's circle problem). $|N(R) - \pi R^2| = O_\epsilon(R^{\frac{1}{2}+\epsilon})$.

Exercise 4.5. Prove that there is a constant B such that

$$\sum_{2 \leq n \leq x} \frac{1}{n \log n} = \log \log x + B + O\left(\frac{1}{x \log x}\right).$$

5 Bernoulli polynomials

Definition 5.1. The r th Bernoulli polynomial $B_r(x)$ and the r th Bernoulli number $B_r = B_r(0)$ is given by the power series expansions

$$F(x, t) = \frac{t}{e^t - 1} e^{xt} = \sum_{r=0}^{\infty} B_r(x) \frac{t^r}{r!}, \quad \text{and} \quad \frac{t}{e^t - 1} = \sum_{r=0}^{\infty} B_r \frac{t^r}{r!}.$$

Proposition 5.2.

1. The Bernoulli polynomials are characterized by the recursive relations

$$B_0(x) = 1, \quad B'_r(x) = r B_{r-1}(x), \quad \int_0^1 B_r(x) dx = 0 \quad (r \in \mathbb{N})$$

2. For all $r \in \mathbb{N}$,

$$B_r(x+1) - B_r(x) = r x^{r-1} \quad \text{and} \quad B_r(1-x) = (-1)^r B_r(x).$$

In particular, $B_r(0) = B_r(1)$ for all $r \geq 2$, and $b_r = 0$ if $r > 1$ is odd.

Proof. (1)

$$F(x, 0) = 1, \quad \frac{\partial}{\partial x} F(x, t) = t F(x, t),$$

$$\int_0^1 F(x, t) dx = \frac{F(1, t) - F(0, t)}{t} = 1.$$

(2)

$$F(x+1, t) - F(x, t) = t e^{xt} = \frac{\partial e^{xt}}{\partial x} = \sum_{r=0}^{\infty} r x^{r-1} \frac{t^r}{r!}.$$

$$F(x, -t) = \frac{-t}{e^{-t} - 1} e^{-xt} = \frac{t}{e^t - 1} e^{(1-x)t} = F(1-x, t). \quad \square$$

Corollary 5.3. The function $B_r(\{x\})$ is continuous if $r \geq 2$.

Exercise 5.4. Prove that $B_1(x) = x - \frac{1}{2}$, $B_2(x) = x^2 - x + \frac{1}{6}$.

Exercise 5.5 (Faulhaber's formula). For all $r, N \in \mathbb{N}$,

$$\sum_{m=0}^{N-1} m^{r-1} = \frac{B_r(N) - B_r}{r}.$$

6 Euler–Maclaurin formula

Definition 6.1. Let $f, g : [y, x] \rightarrow \mathbb{C}$ be functions. The Riemann–Stieltjes integral of f with respect to g is given by

$$\int_y^x f(t) dg(t) = \lim \sum_{i=1}^n f(c_i)(g(x_i) - g(x_{i-1})).$$

where the limit is taken over all partitions $y = x_0 < x_1 < \cdots < x_n = x$ and choices of points $c_i \in [x_{i-1}, x_i]$, as $n \rightarrow \infty$ and $\sup_i (x_i - x_{i-1}) \rightarrow 0$.

Exercise 6.2.

1. If g is continuous and piecewise C^1 , then $\int_y^x f(t) dg(t) = \int_y^x f(t)g'(t) dt$.
2. If f is continuous and g is of *bounded variation*, i.e., $\sup \sum_{i=1}^n |g(x_i) - g(x_{i-1})| < \infty$ where the supremum runs over all finite partitions $y = x_0 < x_1 < \cdots < x_n = x$, then

$$\int_y^x f dg = [fg]_y^x - \int_y^x g df.$$

3. If a is an arithmetic function and f is continuous on $[y, x] \subseteq \mathbb{R}_{\geq 0}$, then

$$\sum_{y < n \leq x} a_n f(n) = \int_y^x f(t) dS_a(t).$$

Theorem 6.3 (Euler–Maclaurin formula). *Let $f \in C^{k+1}([a, b])$, $a, b \in \mathbb{Z}$.*

$$\begin{aligned} \sum_{a < n \leq b} f(n) &= \int_a^b f(t) dt + \sum_{r=0}^k \frac{(-1)^{r+1} B_{r+1}}{(r+1)!} (f^{(r)}(b) - f^{(r)}(a)) \\ &\quad + \frac{(-1)^k}{(k+1)!} \int_a^b B_{k+1}(\{t\}) f^{(k+1)}(t) dt. \end{aligned}$$

Proof. Since $B_1(\{t\}) = \{t\} - \frac{1}{2}$, by Euler’s formula and integration by parts,

$$\begin{aligned} \sum_{a < n \leq b} f(n) &= \int_a^b f(t) d[t] = \int_a^b f(t) dt - \int_a^b f(t) dB_1(\{t\}) \\ &= \int_a^b f(t) dt - B_1 \cdot (f(b) - f(a)) + \int_a^b B_1(\{t\}) f'(t) dt. \end{aligned}$$

Iteratively integrate by parts $\int_a^b B_1(\{t\}) f'(t) dt = \int_a^b f'(t) dB_2(\{t\})$. □

7 Applications of Euler–Maclaurin

Example 7.1. There exists a constant c such that

$$\sum_{n \leq x} \log n = x \log x - x + \frac{1}{2} \log x + c + O\left(\frac{1}{x}\right).$$

Proof. Applying the Euler–Maclaurin formula with $k = 1$, we have

$$\begin{aligned} \sum_{n \leq x} \log n &= \int_1^x \log t \, dt - B_1(\log x - \log 1) \\ &\quad + \frac{B_2}{2} \left(\frac{1}{x} - 1 \right) + \frac{1}{2} \int_1^x \frac{B_2(\{t\})}{t^2} \, dt \\ &= x \log x - x + \frac{1}{2} \log x \\ &\quad + \left(1 - \frac{B_2}{2} + \frac{1}{2} \int_1^\infty \frac{B_2(\{t\})}{t^2} \, dt \right) + O\left(\frac{1}{x}\right). \end{aligned} \quad \square$$

Example 7.2.

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + \frac{1}{2x} - \frac{1}{12x^2} + O\left(\frac{1}{x^3}\right).$$

Proof. Apply the Euler–Maclaurin formula with $k = 2$. We have

$$\begin{aligned} \sum_{n \leq x} \frac{1}{n} &= 1 + \int_1^x \frac{dt}{t} - B_1\left(\frac{1}{x} - 1\right) \\ &\quad + \frac{B_2}{2} \left(-\frac{1}{x^2} + 1 \right) + \frac{1}{6} \int_1^x B_3(\{t\}) \frac{2}{t^3} \, dt \\ &= \log x + \gamma + \frac{1}{2x} - \frac{1}{12x^2} + O\left(\frac{1}{x^3}\right) \end{aligned}$$

where we use the fact that we already know $\sum_{n \leq x} \frac{1}{n} - \log n - \gamma = O(1/x)$. $\square$

8 Riemann zeta function

Remark 8.1. See Lemma 25.2 for analyticity of functions defined by a sequence or series of analytic functions.

Definition 8.2. The Riemann zeta function is the analytic function defined for $\Re(s) > 1$ by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Example 8.3. If $s \neq 1$, then

$$\begin{aligned} \sum_{n \leq x} \frac{1}{n^s} &= 1 + \left[\frac{t^{1-s}}{1-s} \right]_1^x - \left[\frac{\{t\}}{t^s} \right]_1^x - s \int_1^x \frac{\{t\}}{t^{s+1}} dt \\ &= \frac{s}{s-1} + \frac{x^{1-s}}{1-s} - \{x\}x^{-s} - s \int_1^x \frac{\{t\}}{t^{s+1}} dt. \end{aligned}$$

If moreover $\Re(s) > 1$, then

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{t\}}{t^{s+1}} dt - \left(-\frac{x^{1-s}}{1-s} - s \int_x^{\infty} \frac{\{t\}}{t^{s+1}} dt + \{x\}x^{-s} \right).$$

In particular, if $\Re(s) > 1$ then $\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges absolutely and

$$\boxed{\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx.}$$

The right hand side, hence $\zeta(s)$, is an analytic function of s for $\Re(s) > 0$ and $s \neq 1$, with a simple pole at $s = 1$ with $\lim_{s \rightarrow 1} (s-1)\zeta(s) = 1$.

Corollary 8.4. If $\Re(s) > 1$, then

$$\sum_{n > x} \frac{1}{n^s} = -\frac{x^{1-s}}{1-s} - s \int_x^{\infty} \frac{\{t\}}{t^{s+1}} dt + \{x\}x^{-s} = O_s(x^{1-\Re(s)}).$$

Remark 8.5. We will later see that $\zeta(s)$ extends to an meromorphic continuation to the entire complex plane that is analytic

9 Basel problem

Remark 9.1. We have $e^{iz} = \cos z + i \sin z$, so that

$$\cos z = (e^{iz} + e^{-iz})/2, \quad \sin z = (e^{iz} - e^{-iz})/(2i)$$

which gives us, writing $t = 2iz$,

$$\begin{aligned} z \cot z &= z \frac{\cos z}{\sin z} = iz \frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}} = \frac{t e^t + 1}{2 e^t - 1} \\ &= \frac{t}{2} + \frac{t}{e^t - 1} = \sum_{m=0}^{\infty} B_{2m} \frac{t^{2m}}{(2m)!} = \sum_{m=0}^{\infty} (-1)^m 2^{2m} B_{2m} \frac{z^{2m}}{(2m)!}. \end{aligned}$$

Therefore,

$$\cot z - \frac{1}{z} = \sum_{m=1}^{\infty} \frac{(-1)^m 2^{2m} B_{2m}}{(2m)!} z^{2m-1}.$$

Theorem 9.2 (Euler).

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad \text{and} \quad \zeta(2k) = (-1)^{k+1} \frac{B_{2k} (2\pi)^{2k}}{2(2k)!} \quad \forall k \in \mathbb{N}.$$

Proof. We start with Euler's product expansion

$$\frac{\sin z}{z} = \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{(k\pi)^2} \right).$$

Taking the logarithmic derivative of both sides, we get

$$\cot z - \frac{1}{z} = - \sum_{k=1}^{\infty} \frac{2z/(k\pi)^2}{1 - z^2/(k\pi)^2} = - \frac{2}{z} \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} \frac{z^{2m}}{(k\pi)^{2m}} = - \frac{2}{z} \sum_{m=1}^{\infty} \frac{\zeta(2m)}{\pi^{2m}} z^{2m}.$$

Compare with the previous expansion of left hand side. □

Remark 9.3. Apéry (1978) proved that $\zeta(3)$ is irrational.

Exercise 9.4. The multiple zeta function is

$$\zeta(s_1, \dots, s_k) = \sum_{n_1 > \dots > n_k > 0} \frac{1}{n_1^{s_1} \cdots n_k^{s_k}}.$$

Show that $\zeta(\underbrace{2, \dots, 2}_k) = \pi^{2k}/(2k+1)!$.

10 Euler's and Stirling's formulas

Proposition 10.1 (Euler).

$$\boxed{\frac{\sin z}{z} = \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{(k\pi)^2}\right) = \prod_{k=1}^{\infty} \left(1 - \frac{z}{k\pi}\right) \left(1 + \frac{z}{k\pi}\right)}.$$

Proof sketch. Write $N \in \mathbb{N}$. One can show using trigonometric identities

$$\frac{\sin((2N+1)z)}{(2N+1)\sin z} = \prod_{k=1}^N \left(1 - \frac{\sin^2 z}{\sin^2(k\pi/(2N+1))}\right).$$

Thus

$$\frac{\sin(z)}{(2N+1)\sin(z/(2N+1))} = \prod_{k=1}^N \left(1 - \frac{\sin^2(z/(2N+1))}{\sin^2(k\pi/(2N+1))}\right).$$

Letting $N \rightarrow \infty$, using $\lim_{n \rightarrow \infty} \sin(x/n)/(x/n) = 1$ LHS converges to $(\sin z)/z$. Similarly, each term on RHS tends to $1 - z^2/(k\pi)^2$. To conclude local uniform convergence, one checks $\forall M > 0 \exists K > 0$ such that, if $|z| \leq M$, then $|1 - \prod_{k=K+1}^N (1 - \sin^2 z / \sin^2(k\pi/(2N+1)))| \leq \frac{M^2}{4} \sum_{k>K} \frac{1}{k^2}$. $\square$

Corollary 10.2 (Wallis product).

$$\boxed{\frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{2k}{2k-1} \frac{2k}{2k+1}}. \quad \square$$

Corollary 10.3 (Stirling's formula).

$$N! \sim \sqrt{2\pi N} \left(\frac{N}{e}\right)^N.$$

Proof. Exponentiating Example 7.1, there exists a constant $C > 0$ such that $N! \sim C\sqrt{N} (N/e)^N$. So

$$\begin{aligned} p_N &= \prod_{k=1}^N \frac{2k}{2k-1} \frac{2k}{2k+1} = \prod_{k=1}^N \frac{(2k)^4}{(2k-1)(2k) \cdot (2k)(2k+1)} \\ &= \frac{2^{4N}}{2N+1} \frac{(N!)^4}{[(2N)!]^2} \sim \frac{2^{4N}}{2N+1} \frac{C^4 N^2 (N/e)^{4N}}{C^2 (2N)(2N/e)^{4N}} = \frac{C^2 N^2}{(2N+1)(2N)}. \end{aligned}$$

Using $\lim_{N \rightarrow \infty} p_N = \pi/2$, we get $C = \sqrt{2\pi}$ as desired. $\square$

11 Dirichlet convolutions

Definition 11.1. The Dirichlet convolution (or product) $f * g$ of two arithmetic functions f and g is

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d), \quad n \in \mathbb{N}.$$

Proposition 11.2. $*$ is associative and commutative, with identity δ_1 .

Proof.

$$(f * g)(n) = \sum_{ab=n} f(a)g(b) = (g * f)(n)$$

$$((f * g) * h)(n) = \sum_{abc=n} f(a)g(b)h(c) = (f * (g * h))(n)$$

$$(\delta_1 * f)(n) = f(n) = (f * \delta_1)(n). \quad \square$$

Example 11.3. $d(n) = \sum_{d|n} 1 = d = 1 * 1$. More generally,

$$d_k(n) = \#\{(d_1, \dots, d_k) \in \mathbb{N}^k : d_1 \cdots d_k = n\} = 1^{*k}(n).$$

Proposition 11.4. Let f and g be arithmetic functions. Then

$$S_{f*g}(x) = \sum_{n \leq x} f(n)S_g(x/n) = \sum_{n \leq x} g(n)S_f(x/n).$$

Proof.

$$S_{f*g}(x) = \sum_{n \leq x} \sum_{ab=n} f(a)g(b) = \sum_{ab \leq x} f(a)g(b) = \sum_{a \leq x} f(a) \sum_{b \leq x/a} g(b). \quad \square$$

Corollary 11.5. If f is an arithmetic function,

$$S_{f*1}(x) = \sum_{n \leq x} \sum_{d|n} f(d) = \sum_{n \leq x} f(n) \left[\frac{x}{n} \right] = \sum_{n \leq x} S_f\left(\frac{x}{n}\right). \quad \square$$

12 Summatory divisor functions

Recall. The divisor function is $\sigma_s(n) = \sum_{d|n} d^s$.

Example 12.1.

$$\sum_{n \leq x} d(n) = \sum_{n \leq x} \sigma_0(x) = x \log x + O(x).$$

Proof.

$$\sum_{n \leq x} d(n) = S_{1*1}(x) = \sum_{n \leq x} \left[\frac{x}{n} \right] = \sum_{n \leq x} \frac{x}{n} + O(x) = x \log x + O(x). \quad \square$$

Example 12.2.

$$\sum_{n \leq x} \sigma_1(n) = \frac{\zeta(2)}{2} x^2 + O(x \log x) \quad \text{and} \quad \sum_{n \leq x} \sigma_{-1}(n) = \zeta(2)x + O(\log x).$$

Proof.

$$\begin{aligned} \sum_{n \leq x} \sigma_1(n) &= \sum_{n \leq x} S_{\text{id}}\left(\frac{x}{n}\right) = \sum_{n \leq x} \sum_{m \leq x/n} m = \sum_{n \leq x} \left(\frac{1}{2} \left(\frac{x}{n}\right)^2 + O\left(\frac{x}{n}\right) \right) \\ &= \frac{1}{2} x^2 \sum_{n \leq x} \frac{1}{n^2} + O(x \log x) = \frac{1}{2} (\zeta(2) + O(x^{-1})) x^2 + O(x \log x). \end{aligned}$$

$$\begin{aligned} \sum_{n \leq x} \sigma_{-1}(n) &= \sum_{n \leq x} \frac{1}{n} \left[\frac{x}{n} \right] = x \sum_{n \leq x} \frac{1}{n^2} + O\left(\sum_{n \leq x} \frac{1}{n}\right) \\ &= x(\zeta(2) + O(x^{-1})) + O(\log x). \quad \square \end{aligned}$$

Exercise 12.3.

$$\sum_{n \leq x} \sigma_\alpha(n) = \frac{\zeta(\alpha+1)}{\alpha+1} x^{\alpha+1} + O(x^{\max(1, \alpha)}) \quad \text{if } 1 \neq \alpha > 0.$$

$$\sum_{n \leq x} \sigma_\alpha(n) = \zeta(1-\alpha) + O(x^{\max(0, \alpha+1)}) \quad \text{if } -1 \neq \alpha < 0.$$

13 Hyperbola method

Proposition 13.1 (Dirichlet's hyperbola method). *Let f and g be arithmetic functions. If $x = yz$ with $y, z > 0$, then*

$$S_{f*g}(x) = \sum_{a \leq y} f(a)S_g(x/a) + \sum_{b \leq z} S_f(x/b)g(b) - S_f(y)S_g(z).$$

Proof. Define

$$\begin{aligned} R &= \{(a, b) \in \mathbb{N}^2 : ab \leq x\}, \\ R_1 &= \{(a, b) \in \mathbb{N}^2 : ab \leq x, a \leq y\}, \\ R_2 &= \{(a, b) \in \mathbb{N}^2 : ab \leq x, b \leq z\}. \end{aligned}$$

Then $R = R_1 \cup R_2$ and $R_1 \cap R_2 = \mathbb{N}^2 \cap ([1, y] \times [1, z])$. We have

$$S_{f*g}(x) = \sum_{(a,b) \in R} f(a)g(b) = \sum_{R_1} + \sum_{R_2} - \sum_{R_1 \cap R_2}$$

and computing each sum gives the result. □

Theorem 13.2 (Dirichlet).

$$\sum_{n \leq x} d(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}).$$

Proof. Apply the hyperbola method with $y = z = \sqrt{x}$. Then

$$\begin{aligned} \sum_{n \leq x} d(n) &= S_{1*1}(x) = \sum_{a \leq \sqrt{x}} \left[\frac{x}{a} \right] + \sum_{b \leq \sqrt{x}} \left[\frac{x}{b} \right] - [\sqrt{x}]^2 \\ &= 2 \left(\sum_{n \leq \sqrt{x}} \frac{x}{n} + O(\sqrt{x}) \right) - x + O(\sqrt{x}) \\ &= 2x \left(\log \sqrt{x} + \gamma + O\left(\frac{1}{\sqrt{x}}\right) \right) - x + O(\sqrt{x}) \\ &= x \log x + (2\gamma - 1)x + O(\sqrt{x}). \end{aligned} \quad \square$$

Problem 13.3 (Dirichlet's divisor problem). Find infimum $\theta > 0$ such that

$$\sum_{n \leq x} d(n) = x \log x + (2\gamma - 1)x + O(x^\theta).$$

14 Dirichlet derivative

Definition 14.1. The *Dirichlet derivative* f' of $f : \mathbb{N} \rightarrow \mathbb{C}$ is given by

$$f'(n) = -f(n) \log n.$$

Exercise 14.2. We have $(f * g)' = f' * g + f * g'$.

Definition 14.3. The von Mangoldt function is given by

$$\Lambda(n) = \begin{cases} \log p & n = p^k \text{ for some prime } p \text{ and } k \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 14.4.

1. We have $1 * \Lambda = -1'$. In other words,

$$\sum_{d|n} \Lambda(d) = \log n.$$

2. We have

$$\sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right] = \log([x]!) = \sum_{n \leq x} \log n.$$

3. (Legendre's identity) $[x]! = \prod_{p \leq x} p^{\alpha(p,x)}$ where $\alpha(p, n) = \sum_{m=1}^{\infty} \left[\frac{x}{p^m} \right]$.

Proof. First,

$$\sum_{d|n} \Lambda(d) = \sum_p \sum_{k=1}^{v_p(n)} \log p = \sum_{p|n} v_p(n) \log p = \log \left(\prod_p p^{v_p(n)} \right) = \log n.$$

Next,

$$\log([x]!) = \sum_{n \leq x} \log n = S_{1*\Lambda}(x) = \sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right].$$

Legendre's identity follows from the fact that

$$\sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right] = \sum_p \sum_{m=1}^{\infty} \log p \left[\frac{x}{p^m} \right].$$

□

15 Multiplicative functions

Definition 15.1. Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function.

1. f is *multiplicative* if $f(1) = 1$ and

$$f(ab) = f(a)f(b) \quad (*)$$

for all $a, b \in \mathbb{N}$ with $(a, b) = 1$.

2. f is *completely multiplicative* if moreover $(*)$ holds for all $a, b \in \mathbb{N}$.

Remark 15.2.

1. f is multiplicative iff $f(1) = 1$ and $f(n) = \prod_p f(p^{v_p(n)})$ for all $n \in \mathbb{N}$.
2. If f and g are multiplicative, $f = g$ iff $f(p^k) = g(p^k)$ for all $p \in \mathbb{P}$, $k \in \mathbb{N}$.

Proposition 15.3. *If f and g are multiplicative, then $f * g$ is multiplicative.*

Proof. Clearly $(f * g)(1) = f(1)g(1) = 1$. Write $D(n) = \{d \in \mathbb{N} : d \mid n\}$. Using the natural bijection $D(n) \simeq \prod_p D(p^{v_p(n)})$,

$$\begin{aligned} (f * g)(n) &= \sum_{d \mid n} f(d)g(n/d) = \sum_{d \mid n} \prod_p f(p^{v_p(d)})g(p^{v_p(n)-v_p(d)}) \\ &= \prod_p \left(\sum_{d \mid p^{v_p(n)}} f(p^{v_p(d)})g(p^{v_p(n)-v_p(d)}) \right) = \prod_p (f * g)(p^{v_p(n)}). \quad \square \end{aligned}$$

Corollary 15.4. *If $f : \mathbb{N} \rightarrow \mathbb{C}$ is multiplicative, so is $(1 * f)(n) = \sum_{d \mid n} f(d)$.*

Example 15.5. The divisor function $\sigma_s(n) = \sum_{d \mid n} d^s$ is multiplicative.

Example 15.6. The Liouville function is $\lambda(n) = (-1)^{\Omega(n)}$ is multiplicative.

$$(1 * \lambda)(n) = \sum_{d \mid n} \lambda(d) = \delta_{\square}(n).$$

Proof. Suffice to check $(1 * \lambda)(p^k) = \sum_{0 \leq \ell \leq k} \lambda(p^\ell) = \delta_{2\mathbb{Z}}(k) = \delta_{\square}(p^k)$. $\square$

16 Dirichlet inverse

Proposition 16.1. *Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function. TFAE:*

1. $f(1) \in \mathbb{C}^*$.
2. *There exists $g : \mathbb{N} \rightarrow \mathbb{C}$, called Dirichlet inverse of f , such that $f * g = \delta_1$.*

Remark 16.2. If such g exists then it is unique and denoted f^{-1} .

Proof. Clearly (2) implies (1) since $f(1)g(1) = 1$. Conversely, let $g(1) = f(1)^{-1}$ and define $g(n)$ for $1 \neq n \in \mathbb{N}$ by the recursive (on $d(n)$ say) formula

$$g(n) = -\frac{1}{f(1)} \sum_{d|n, d \neq n} g(d)f(n/d).$$

Then we have $(f * g)(n) = \delta_1(n)$ for all $n \in \mathbb{N}$. □

Proposition 16.3. *If $f : \mathbb{N} \rightarrow \mathbb{C}$ is multiplicative, then f^{-1} is multiplicative.*

Proof. Let $g : \mathbb{N} \rightarrow \mathbb{C}$ be the unique multiplicative function such that for all $p \in \mathbb{P}$ and $k \in \mathbb{N}$ the recursive formula

$$g(p^k) = - \sum_{0 \leq \ell < k} f(p^{k-\ell})g(p^\ell)$$

holds. Then $(f * g)(p^k) = 0$ for all $p \in \mathbb{P}$ and $k \in \mathbb{N}$. Since $f * g$ is multiplicative, we deduce $f * g = \delta_1$. Hence $f^{-1} = g$ is multiplicative. □

Example 16.4. Let $\lambda(n) = (-1)^{\Omega(n)}$. Then $\lambda^{-1} = \delta_{\text{squarefree}}$.

Proof. For $p \in \mathbb{P}$ and $k \in \mathbb{N}$, recursive formula gives

$$\lambda^{-1}(p) = -\lambda(p) = 1, \quad \lambda^{-1}(p^2) = -\lambda(p^2)\lambda^{-1}(1) - \lambda(p)\lambda^{-1}(p) = 0,$$

and inductively $\lambda^{-1}(p^k) = -\lambda(p^k) - \lambda(p^{k-1})\lambda^{-1}(p) = 0$ for all $k \geq 3$. □

17 Möbius function

Recall. The Möbius function is

$$\mu(n) = \begin{cases} (-1)^{\omega(n)} & n \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 17.1. *We have $\mu * 1 = \delta_1$. Explicitly, for any $n \in \mathbb{N}$*

$$\boxed{\sum_{d|n} \mu(d) = \delta_1(n).}$$

Proof. Since LHS is multiplicative, it suffices to show LHS is zero for $n = p^k$ with $p \in \mathbb{P}$, $k \in \mathbb{N}$. But

$$\sum_{d|p^k} \mu(d) = \sum_{0 \leq \ell \leq k} \mu(p^\ell) = \mu(1) + \mu(p) = 1 - 1 = 0. \quad \square$$

Corollary 17.2. *If $x \geq 1$, then*

$$\boxed{\sum_{n \leq x} \mu(n) \left[\frac{x}{n} \right] = S_{1*\mu}(x) = 1.}$$

Corollary 17.3. *For all $x \geq 1$ we have*

$$\left| \sum_{n \leq x} \frac{\mu(n)}{n} \right| \leq 1.$$

Proof. The statement is clear if $x < 2$, so assume $x \geq 2$. Then

$$x \left| \sum_{n \leq x} \mu(n) \frac{x}{n} \right| = \left| \sum_{n \leq x} \mu(n) \left[\frac{x}{n} \right] + \sum_{n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} \right| < 1 + \{x\} + ([x] - 1) = x. \quad \square$$

Lemma 17.4. *If $\Re(s) > 1$, then*

$$\boxed{\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}.$$

Proof. Note that the LHS converges absolutely for $\Re(s) > 1$, and by Fubini

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \sum_{m=1}^{\infty} \frac{1}{m^s} \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \sum_{N=1}^{\infty} \frac{\sum_{mn=N} \mu(n)}{N^s} = 1. \quad \square$$

18 Möbius inversion

Theorem 18.1 (Möbius inversion). *Let $f, g : \mathbb{N} \rightarrow \mathbb{C}$ be arithmetic.*

$$g(n) = \sum_{d|n} f(d) \quad \forall n \in \mathbb{N} \quad \Longleftrightarrow \quad f(n) = \sum_{d|n} g(d) \mu(n/d) \quad \forall n \in \mathbb{N}.$$

Proof. $g = 1 * f \Longleftrightarrow f = \mu * g.$ □

Lemma 18.2. *If f is multiplicative, then*

$$\sum_{d|n} \mu(d) f(d) = \prod_{p|n} (1 - f(p)).$$

Proof. Since both sides are multiplicative, suffice to check formula for $n = p^k$ with $p \in \mathbb{P}$, $k \in \mathbb{N}$. But

$$\sum_{d|p^k} \mu(d) f(d) = \mu(1) f(1) + \mu(p) f(p) = 1 - f(p). \quad \square$$

Definition 18.3. Euler's totient function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is defined to be

$$\varphi(n) = \#\{1 \leq a \leq n : (a, n) = 1\} = \#(\mathbb{Z}/n\mathbb{Z})^*.$$

Proposition 18.4. *We have $1 * \varphi = \text{id}$, i.e., for each $n \in \mathbb{N}$,*

$$\sum_{d|n} \varphi(d) = n$$

Proof. Write $A_d(n) = \{1 \leq a \leq n : (a, n) = d\}$, so that $\varphi(n) = \#A_1(n)$. Then $\{1, \dots, n\} = \coprod_{d|n} A_d(n)$ and $A_d(n) \simeq A_1(n/d)$ under $a \mapsto a/d$. □

Corollary 18.5. *We have*

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Proof. By Möbius inversion $\varphi = \mu * \text{id}$ and

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d} = n \prod_{p|n} \left(1 - \frac{1}{p}\right). \quad \square$$

19 Summatory totient function

Theorem 19.1.

$$\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2} x^2 + O(x \log x).$$

Proof. From $\varphi = \mu * \text{id}$ we get

$$\begin{aligned} \sum_{n \leq x} \varphi(n) &= S_{\mu * \text{id}}(x) = \sum_{n \leq x} \mu(n) S_{\text{id}}\left(\frac{x}{n}\right) = \sum_{n \leq x} \mu(n) \left(\frac{1}{2} \left(\frac{x}{n}\right)^2 + O\left(\frac{x}{n}\right) \right) \\ &= \sum_{n \leq x} \frac{\mu(n)}{2} \frac{x^2}{n^2} + O\left(\sum_{n \leq x} \frac{x}{n}\right) = \frac{x^2}{2} \left(\frac{1}{\zeta(2)} + O(x^{-1}) \right) + O(x \log x) \\ &= \frac{1}{2\zeta(2)} x^2 + O(x \log x) = \frac{3}{\pi^2} x^2 + O(x \log x). \quad \square \end{aligned}$$

Exercise 19.2. The probability that two randomly chosen integers are coprime is

$$\lim_{x \rightarrow \infty} \frac{\{(a, b) \in \mathbb{Z}^2 : |a|, |b| \leq x, (a, b) = 1\}}{\{(a, b) \in \mathbb{Z}^2 : |a|, |b| \leq x\}} = \frac{6}{\pi^2}.$$

Example 19.3. Probability that a randomly chosen $n \in \mathbb{N}$ is squarefree is

$$\lim_{x \rightarrow \infty} \frac{\{n \leq x : n \text{ squarefree}\}}{\{n \leq x\}} = \frac{6}{\pi^2}.$$

Proof. Each $n \in \mathbb{N}$ can be written uniquely as $a^2 b$ where $a \in \mathbb{N}$ and b is squarefree. Thus, we have $\delta_{\text{sqfree}} * \delta_{\square} = 1$. Note that $\delta_{\square}^{-1}(n) = \mu(\sqrt{n})$. So

$$\delta_{\text{sqfree}}(n) = \sum_{d^2 | n} \mu(d).$$

Thus,

$$\begin{aligned} S_{\text{sqfree}}(x) &= \sum_{n \leq x} \sum_{d^2 | n} \mu(d) = \sum_{d \leq \sqrt{x}} \sum_{n \leq x/d^2} \mu(d) \\ &= \sum_{d \leq \sqrt{x}} \mu(d) \left[\frac{x}{d^2} \right] = \frac{x}{\zeta(2)} + O(\sqrt{x}). \end{aligned}$$

dividing by x and letting $x \rightarrow \infty$, we get the result. $\square$

20 Formal Dirichlet series

Definition 20.1. The formal Dirichlet series associated to arithmetic function $a = (a_n)_{n \in \mathbb{N}}$ is given by

$$\mathbb{D}(a, s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

Remark 20.2. Formal Dirichlet series form a commutative ring under

$$\begin{aligned} \mathbb{D}(a, s) + \mathbb{D}(b, s) &= \sum_{n=1}^{\infty} \frac{a_n + b_n}{n^s} = \mathbb{D}(a + b, s), \\ \mathbb{D}(a, s)\mathbb{D}(b, s) &= \sum_{m, n \geq 1} \frac{a_m b_n}{(mn)^s} = \sum_{N=1}^{\infty} \frac{(a * b)_N}{N^s} = \mathbb{D}(a * b, s). \end{aligned}$$

with identity $1 = \mathbb{D}(\delta_1, s)$. If $a = (a_n)$ satisfies $a_1 \in \mathbb{C}^*$, then $\mathbb{D}(a, s)$ is invertible and $\mathbb{D}(a, s)^{-1} = \mathbb{D}(a^{-1}, s)$. The formal derivative

$$\mathbb{D}'(a, s) = \sum_{n=1}^{\infty} (-a_n \log n) / n^s = \mathbb{D}(a', s)$$

satisfies $(\mathbb{D}(a, s)\mathbb{D}(b, s))' = \mathbb{D}'(a, s)\mathbb{D}(b, s) + \mathbb{D}(a, s)\mathbb{D}'(b, s)$.

Example 20.3. If f is an arithmetic function and $g(n) = \sum_{d|n} f(d)$,

$$\mathbb{D}(g, s) = \mathbb{D}(f, s)\zeta(s).$$

Here, we write $\zeta(s) = \mathbb{D}(1, s)$.

Definition 20.4. Given a multiplicative function $a = (a_n)_{n \in \mathbb{N}}$, the formal Euler product expansion of its formal Dirichlet series is

$$\mathbb{D}(a, s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} = \prod_{p \in \mathbb{P}} \left(1 + \frac{a_p}{p^s} + \frac{a_{p^2}}{p^{2s}} + \cdots \right).$$

If a is completely multiplicative, then we write

$$\mathbb{D}(a, s) = \prod_{p \in \mathbb{P}} \left(1 + \frac{a_p}{p^s} + \frac{a_p^2}{p^{2s}} + \cdots \right) = \prod_{p \in \mathbb{P}} \left(1 - \frac{a_p}{p^s} \right)^{-1}.$$

Remark 20.5. We may view $\mathbb{D}(a, s)$ as a function of a complex variable s where the series converges. We will write $D(a, s)$ to denote the function.

21 Examples of formal Dirichlet series

Example 21.1. Formally, we have the following.

1. $\mathbb{D}(1, s) = \zeta(s)$.
2. $\mathbb{D}(\text{id}^\alpha, s) = \sum_{n=1}^{\infty} \frac{n^\alpha}{n^s} = \zeta(s - \alpha)$.
3. $\mathbb{D}(d_k, s) = \mathbb{D}(1^{*k}, s) = \zeta(s)^k$.
4. $\mathbb{D}(\sigma_\alpha, s) = \mathbb{D}(1 * \text{id}^\alpha, s) = \zeta(s)\zeta(s - \alpha)$.
5. $\mathbb{D}(\Lambda, s) = \mathbb{D}(-1' * 1^{-1}, s) = -\zeta'(s)/\zeta(s)$.
6. $\mathbb{D}(\mu, s) = \mathbb{D}(1^{-1}, s) = \zeta(s)^{-1}$.
7. $\mathbb{D}(\varphi, s) = \mathbb{D}(\text{id} * \mu, s) = \zeta(s - 1)/\zeta(s)$.

Exercise 21.2. Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function. Then

$$\mathbb{D}(\lambda, s) = \frac{\zeta(2s)}{\zeta(s)}.$$

Exercise 21.3. Let $k \in \mathbb{N}$. Define $q_k(n) = 1$ if n is k th power free and 0 otherwise. Then formally

$$\sum_{n=1}^{\infty} \frac{q_k(n)}{n^s} = \frac{\zeta(s)}{\zeta(ks)}.$$

Exercise 21.4. For any $a, b \in \mathbb{C}$, formally

$$\boxed{\sum_{n=1}^{\infty} \frac{\sigma_a(n)\sigma_b(n)}{n^s} = \frac{\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)}{\zeta(2s-a-b)}}, \quad \boxed{\sum_{n=1}^{\infty} \frac{d(n)^2}{n^s} = \frac{\zeta(s)^4}{\zeta(2s)}}.$$

Proof. Let $n = p^k$ where $p \in \mathbb{P}$ and $k \in \mathbb{N}$. Then

$$\sum_{k=0}^{\infty} \frac{\sigma_a(p^k)\sigma_b(p^k)}{p^{ks}} = \sum_{k=0}^{\infty} \frac{1}{p^{ks}} \left(\sum_{\ell=0}^k p^{a\ell} \right) \left(\sum_{\ell=0}^k p^{b\ell} \right) = \sum_{k=0}^{\infty} \frac{p^{a(k+1)} - 1}{p^a - 1} \frac{p^{b(k+1)} - 1}{p^b - 1} \frac{1}{p^{ks}}.$$

Use Ramanujan's identity

$$\sum_{n=0}^{\infty} \left(\frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} \right) \left(\frac{\gamma^{n+1} - \delta^{n+1}}{\gamma - \delta} \right) T^n = \frac{1 - \alpha\beta\gamma\delta T^2}{(1 - \alpha\gamma T)(1 - \alpha\delta T)(1 - \beta\gamma T)(1 - \beta\delta T)}$$

and the Euler product expansion to conclude the desired result. $\square$

22 Absolute convergence

It is customary to write $\sigma = \Re s$, $t = \Im s$, and $s = \sigma + it$.

Definition 22.1. Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function. Then there exists $\sigma_a = \sigma_a(f) \in [-\infty, \infty]$ called abscissa of absolute convergence such that $\mathbb{D}(f, s)$ converges absolutely for all $\Re s > \sigma_a$ and does not converge absolutely for all $\Re s > \sigma_a$.

Proof. Define $\sigma_a = \sup\{\sigma : \sum_{n=1}^{\infty} |a_n|/n^{\sigma} \text{ diverges}\}$. □

Theorem 22.2 (Uniqueness). *Let $f, g : \mathbb{N} \rightarrow \mathbb{C}$ with $\sigma_a(f), \sigma_a(g) < \infty$. Assume there is a sequence $\{s_k\}$ in $\mathbb{C}$ such that $\lim_{k \rightarrow \infty} \sigma_k = +\infty$ and $D(f, s_k) = D(g, s_k)$ for all $k \in \mathbb{N}$. Then $f = g$.*

Proof. We may assume $g = 0$. Assume there exists a least $N \in \mathbb{N}$ such that $f(N) \neq 0$. We may assume $\sigma_k > c > \sigma_a$. Since $D(f, s_k) = 0$,

$$\begin{aligned} |f(N)| &= \left| N^{s_k} D(f, s_k) - N^s \sum_{n=N+1}^{\infty} \frac{f(n)}{n^{s_k}} \right| = N^{\sigma_k} \left| \sum_{n=N+1}^{\infty} \frac{f(n)}{n^{s_k}} \right| \\ &\leq N^{\sigma_k} \sum_{n=N+1}^{\infty} \frac{|f(n)|}{n^c} n^{c-\sigma_k} \leq \left(\frac{N}{N+1} \right)^{\sigma_k} (N+1)^c D(|f|, c). \end{aligned}$$

Letting $k \rightarrow \infty$, we deduce $f(N) = 0$, a contradiction. □

Corollary 22.3. *Let $f : \mathbb{N} \rightarrow \mathbb{C}$ such that $\sigma_a(f) < \infty$. Then either $f = 0$ or there exists $c \geq \sigma_a(f)$ such that $D(f, s) \neq 0$ for all $\Re s > c$.*

Exercise 22.4. If $\sigma_a(f) < \infty$, then $\lim_{\sigma \rightarrow \infty} D(f, \sigma + it) = f(1)$ uniformly for $t \in \mathbb{R}$.

Proposition 22.5. *Let $f, g : \mathbb{N} \rightarrow \mathbb{C}$. Assume that $\sigma_a(f), \sigma_a(g) < \infty$. Then $\sigma_a(f * g) \leq \max\{\sigma_a(f), \sigma_a(g)\}$ and*

$$D(f, s)D(g, s) = D(f * g, s)$$

holds for all $\Re s > \max\{\sigma_a(f), \sigma_a(g)\}$.

Proof. By absolute convergence, we can apply Fubini's theorem. □

Example 22.6. $\mathbb{D}(\varphi, s)$ converges absolutely for $\sigma > 2$ where we have $D(\varphi, s) = \zeta(s-1)/\zeta(s)$.

23 Euler products

Theorem 23.1. *Let f be a multiplicative function. Then the Euler product*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots \right)$$

converges absolutely for $\sigma > \sigma_a(f)$. If f is completely multiplicative, then

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(1 - \frac{f(p)}{p^s} \right)^{-1}.$$

Proof. Let $\Re s > \sigma_a(f)$. Let us write $P_s(x) = \prod_{p \leq x} (\sum_{k=0}^{\infty} f(p^k)/p^{ks})$. Then

$$\left| \sum_{n \leq x} \frac{f(n)}{n^s} - P_s(x) \right| \leq \sum_{n > x} \frac{|f(n)|}{n^{\sigma}}.$$

Since $D(f, s)$ is absolutely convergent, the right hand side tends to zero as $x \rightarrow \infty$. So $\lim_{x \rightarrow \infty} P_s(x) = D(f, s)$ pointwise. Absolute convergence comes from the fact that a product of the form $\prod_k (1 + a_k)$ is absolutely convergent if $\sum a_k$ is absolutely convergent, and we have

$$\sum_p \left| \sum_{k=1}^{\infty} \frac{f(p^k)}{p^{ks}} \right| \leq \sum_p \sum_{k=1}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} \leq \sum_{n=1}^{\infty} \frac{|f(n)|}{n^{\sigma}}. \quad \square$$

Example 23.2. We have the Euler product expansion

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s} \right)^{-1}$$

valid for $\sigma > 1$. Since $\zeta(s)$ has a simple pole at $s = 1$, the product on the right hand side must be infinite so there are infinitely many primes!

Example 23.3.

$$\sum_{n=1}^{\infty} \frac{\sigma_{\alpha}(n)}{n^s} = \zeta(s)\zeta(s - \alpha) = \prod_p \frac{1}{(1 - p^{-s})(1 - p^{-(s-\alpha)})}$$

for $\sigma > \max\{1, 1 + \Re(\alpha)\}$.

24 Convergence of Dirichlet series

Theorem 24.1. *Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function. If $\mathbb{D}(f, s)$ converges for some $s_0 \in \mathbb{C}$ then it also converges for all $\Re(s) > \Re(s_0)$. If it diverges for some $s_0 \in \mathbb{C}$ then it also diverges for all $\Re(s) < \Re(s_0)$.*

Proof. The second statement follows from the first. To prove the first, fix $\Re(s) > \Re(s_0)$. By assumption $\mathbb{D}(f, s_0)$ has bounded partial sums in that there exists $M > 0$ with $|\sum_{n \leq x} f(n)/n^{s_0}| \leq M$ for all $x \geq 1$. Write $g(n) = f(n)n^{-s_0}$. Then by partial summation

$$\begin{aligned} \left| \sum_{a < n \leq b} \frac{f(n)}{n^s} \right| &= \left| \sum_{a < n \leq b} \frac{g(n)}{n^{s-s_0}} \right| = \left| \left[\frac{S_g(t)}{t^{s-s_0}} \right]_a^b + (s-s_0) \int_a^b \frac{S_g(t)}{t^{s-s_0+1}} dt \right| \\ &\leq 2Ma^{\sigma_0-\sigma} + |s-s_0|M \left| \frac{b^{\sigma_0-\sigma} - a^{\sigma_0-\sigma}}{\sigma_0-\sigma} \right| \\ &\leq 2M \left(1 + \frac{|s-s_0|}{\sigma-\sigma_0} \right) a^{\sigma_0-\sigma} = Ka^{\sigma_0-\sigma} \end{aligned}$$

where K is independent of a . By Cauchy's criterion, $\mathbb{D}(f, s)$ converges. $\square$

Example 24.2. Above proof shows that if $\sum_{n \leq x} f(n)$ is bounded then $\mathbb{D}(f, s)$ converges for $\Re s > 0$. For example, the Dirichlet series

$$\sum_{n=1}^{\infty} (-1)^n / n^s$$

converges for $\Re s > 0$. It converges absolutely for $\Re s > 1$.

Definition 24.3. Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be an arithmetic function. Then there exists $\sigma_c = \sigma_c(f) \in [-\infty, \infty]$ called abscissa of convergence such that $\mathbb{D}(f, s)$ converges for all $\Re s > \sigma_c$ and diverges for all $\Re s < \sigma_c$.

Proof. Take $\sigma_c = \sup\{\Re(s) : \mathbb{D}(f, s) \text{ diverges}\}$. $\square$

Proposition 24.4. *If $\sigma_c < \infty$, then $\sigma_c \leq \sigma_a \leq \sigma_c + 1$.*

Proof. Let $\Re s_0 > \sigma_c$. Since $\mathbb{D}(f, s_0)$ converges, there exists $A > 0$ such that $|f(n)/n^{-s_0}| \leq A$ for all $n \in \mathbb{N}$. If $\Re(s) > \Re(s_0) + 1$ then $\mathbb{D}(f, s)$ is absolutely convergent since $|f(n)n^{-s}| \leq An^{\sigma-s_0}$. $\square$

25 Analyticity of Dirichlet series

Theorem 25.1. *A Dirichlet series $\mathbb{D}(f, s)$ converges uniformly on every compact subset in the half-plane of convergence $\Re(s) > \sigma_c(f)$ and in particular is analytic there, with*

$$D'(f, s) = D(f', s) = - \sum_{n=1}^{\infty} \frac{f(n) \log n}{n^s} \quad \text{for } \Re(s) > \sigma_c(f).$$

Proof. Let $R = [\alpha, \beta] \times [c, d]$ be a compact rectangle in the complex plane with $\alpha > \sigma_c(f)$. Fix $\alpha > \sigma_0 > \sigma_c(f)$. By arguing as in the proof of Theorem 24.1, there exists $M > 0$ such that for any $s \in R$ we have

$$\left| \sum_{a < n \leq b} \frac{f(n)}{n^s} \right| \leq 2M \left(1 + \frac{|s - s_0|}{\sigma - \sigma_0} \right) a^{\sigma_0 - \sigma} \leq 2M \left(1 + \frac{\max_{s \in R} |s - s_0|}{\alpha - \sigma_0} \right) a^{\sigma_0 - \alpha}.$$

RHS is independent of $s \in R$, so $\mathbb{D}(f, s)$ converges uniformly on $s \in R$. The rest follows by Lemma 25.2 □

Lemma 25.2. *let (f_n) be a sequence of functions analytic on an open subset $S \subseteq \mathbb{C}$. Assume (f_n) converges uniformly on every compact subset of S . Then the pointwise limit $f(z) = \lim_{n \rightarrow \infty} f_n(z)$ is analytic on S and the sequence of derivatives f'_n converges uniformly on every compact subset of S to the derivative f' .*

Proof. For any $a \in S$ we have by Cauchy's integral formula

$$f_n(a) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f_n(z)}{z - a} dz$$

for any compact disk $D \subset S$ with $a \in D^\circ$. By uniform convergence, interchanging limit with integral we get $f(a) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(z)}{z - a} dz$ showing f is analytic. For the derivatives we have

$$f'_n(a) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f_n(z)}{(z - a)^2} dz, \quad f'(a) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(z)}{(z - a)^2} dz$$

which shows $f'_n \rightarrow f'$ uniformly on every compact subset of S . □

26 Summatory bounds and Dirichlet series

Proposition 26.1.

1. Let $\delta \in \mathbb{R}$. If $S_a(x) = O(x^\delta)$, then $\mathbb{D}(a, s)$ converges for $\Re(s) > \delta$ and

$$D(a, s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} = s \int_1^{\infty} \frac{S_a(t)}{t^{s+1}} dt.$$

2. Let $\Re(s_0) > 0$. If $\mathbb{D}(a, s_0)$ converges, then $S_a(x) = o(x^{\Re(s_0)})$.

Proof. (1) We have

$$\begin{aligned} \sum_{n \leq x} \frac{a_n}{n^s} &= a_1 + [S_a(t)t^{-s}]_1^x + s \int_1^x \frac{S_a(t)}{t^{s+1}} dt \\ &= S_a(x)x^{-s} + s \int_1^{\infty} \frac{S_a(t)}{t^{s+1}} dt - s \int_x^{\infty} \frac{S_a(t)}{t^{s+1}} dt \\ &= s \int_1^{\infty} \frac{S_a(t)}{t^{s+1}} dt + O(x^{-(\Re(s)-\delta)}). \end{aligned}$$

(2) Write $b_n = a_n/n^{s_0}$. Since $\sum_{n=1}^{\infty} b_n = B$ converges, writing $r(x) = \sum_{n>x} b_n$ we have

$$\begin{aligned} S_a(N) &= \sum_{n \leq x} b(n)n^{s_0} = b_1 + [S_b(t)t^{s_0}]_1^x + s_0 \int_1^x S_b(t)t^{s_0-1} dx \\ &= b_1 + [(B - r(t))t^{s_0}]_1^x + s_0 \int_1^x (B - r(t))t^{s_0-1} dx \\ &= r(t)x^{s_0} - s_0 \int_1^x r(t)t^{s_0-1} dt + O(1). \end{aligned}$$

Since $r(x) = o(1)$, it suffices to show that $\int_1^x r(t)t^{s_0-1} dt = o(x^{s_0})$. To see this, given $\epsilon > 0$ choose $T > 1$ such that $|r(x)| < \epsilon$ for all $x > T$. Then

$$\left| \int_T^x r(t)t^{s_0-1} dt \right| \leq \frac{\epsilon}{s_0} x^{s_0} + O_{\epsilon, T}(1).$$

Thus for every $\epsilon > 0$ we have $\limsup_{x \rightarrow \infty} (\int_1^x r(t)t^{s_0-1} dt)/x^{s_0} \leq \epsilon$. The result follows. $\square$

Example 26.2. In the case $a_n \equiv 1$ and $\delta = 1$, we recover $\Re(s) > \delta$

$$\zeta(s) = s \int_1^{\infty} \frac{[t]}{t^{s+1}} dt = \left[\frac{s}{1-s} t^{1-s} \right]_1^{\infty} - s \int_1^{\infty} \frac{\{t\}}{t^{s+1}} dt = \frac{s}{s-1} - \int_1^{\infty} \frac{\{t\}}{t^{s+1}} dt.$$

27 Chebyshev functions

Definition 27.1. The first Chebyshev function is

$$\vartheta(x) = \sum_{p \leq x} \log p.$$

The second Chebyshev function is

$$\psi(x) = S_{\Lambda}(x) = \sum_{p^k \leq x} \log p = \sum_{k=1}^{\infty} \vartheta(x^{1/k}).$$

Remark 27.2. Using the trivial inequality $\vartheta(x) \leq x \log x$ we have

$$0 \leq \psi(x) - \vartheta(x) = \sum_{2 \leq k \leq \log_2 x} \vartheta(x^{1/k}) \leq \frac{\log x}{\log 2} \cdot \sqrt{x} \log \sqrt{x} = o(x).$$

Definition 27.3. Let us write $f(x) \asymp g(x)$ if $g(x) \ll f(x) \ll g(x)$.

Theorem 27.4 (Chebyshev). $\vartheta(x) \asymp x$ and $\psi(x) \asymp x$.

Proof. Suffices to show $\vartheta(n) \leq 2n \log 2 \leq \psi(2n+1)$. If $n < p \leq 2n$ and p is prime, then $p \mid \binom{2n}{n}$. Thus

$$\prod_{n < p \leq 2n} p \mid \binom{2n}{n} \leq \sum_{k=0}^{2n} \binom{2n}{k} = 2^{2n}.$$

Taking logarithms, first inequality follows. Next, note that (argument due to M. Nair)

$$I = \int_0^1 x^n (1-x)^n dx = \sum_{k=0}^n \binom{n}{k} (-1)^k \int_0^1 x^{n+k} dx = \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{n+k+1} \in \mathbb{Q}_{>0}$$

and using $x(1-x) \leq 1/4$ for $x \in [0, 1]$ we get $I \leq 2^{-2n}$. On the other hand, $\text{lcm}(1, \dots, n) = e^{\psi(n)}$. Indeed, for any prime p , the p -adic valuation of both sides is $\max\{k \in \mathbb{Z}_{\geq 0} : p^k \leq n\}$. Thus we have $e^{\psi(2n+1)} I \geq 1$ and hence $\psi(2n+1) \geq \log(2^{2n}) = 2n \log 2$. $\square$

28 Equivalent formulations of PNT

Proposition 28.1.

$$\boxed{\vartheta(x) = \pi(x) \log x - \int_2^x \frac{\pi(t)}{t} dt} \quad \text{and} \quad \boxed{\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t(\log t)^2} dt.}$$

Proof. By partial summation, we have

$$\vartheta(x) = \log 2 + \sum_{2 < p \leq x} \log p = \log 2 + [\pi(t) \log t]_2^x + \int_2^x \frac{\pi(t)}{t} dt$$

giving the first part. The second part uses $(1/\log t)' = -1/(t(\log t)^2)$. $\square$

Theorem 28.2. *We have*

$$\boxed{\lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = \lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = \lim_{x \rightarrow \infty} \frac{\pi(x)}{x/\log x}}$$

PROVIDED THAT at least one of the limits exists.

Proof. The equality of first two limits (if either exists) follows from the fact that $|\vartheta(x) - \psi(x)| = o(x)$. Now note that, for all $k \in \mathbb{N}$,

$$\int_2^x \frac{dt}{(\log t)^k} = \int_2^{\sqrt{x}} + \int_{\sqrt{x}}^x = \frac{\sqrt{x} - 2}{(\log 2)^k} + \frac{x - \sqrt{x}}{(\log \sqrt{x})^k} = O\left(\frac{x}{(\log x)^k}\right).$$

Suppose $\lim_{x \rightarrow \infty} \frac{\pi(x)}{x/\log x} = \alpha$. We have $\pi(t)/t = O(1/\log t)$ for $t \geq 2$ so

$$\frac{\vartheta(x)}{x} = \frac{\pi(x)}{x/\log x} - \frac{1}{x} \int_2^x \frac{\pi(t)}{t} dt = \alpha + o(1) + \frac{1}{x} O\left(\int_2^x \frac{1}{\log t} dt\right) = \alpha + o(1).$$

Suppose $\lim \vartheta(x)/x = \alpha$. This implies $\vartheta(t)/t = O(1)$ for $t \geq 2$ so

$$\frac{\pi(x)}{x/\log x} = \frac{\vartheta(x)}{x} + \frac{\log x}{x} \int_2^x \frac{\vartheta(t)}{t(\log t)^2} dt = \alpha + o(1) + \frac{\log x}{x} O\left(\int_2^x \frac{dt}{(\log t)^2}\right)$$

which is $\alpha + o(1)$. $\square$

Remark 28.3. Same proof gives $\vartheta(x) \asymp x$ iff $\psi(x) \asymp x$ iff $\pi(x) \asymp x/\log x$.

Theorem 28.4 (Chebyshev). $\boxed{\pi(x) \asymp x/\log x.}$

29 Logarithmic average

Definition 29.1. The logarithmic average of $a : \mathbb{N} \rightarrow \mathbb{C}$ up to x is

$$\frac{1}{\log x} \sum_{n \leq x} \frac{a_n}{n}.$$

Remark 29.2. The above is asymptotic to $(\sum_{n \leq x} a_n/n)/(\sum_{n \leq x} 1/n)$.

Proposition 29.3. *Let a be an arithmetic function. Let $\alpha \in \mathbb{C}$. Then*

$$\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n \leq x} a_n = \alpha \quad \implies \quad \lim_{x \rightarrow \infty} \frac{1}{\log x} \sum_{n \leq x} \frac{a_n}{n} = \alpha.$$

Proof. By partial summation, we have

$$\sum_{n \leq x} \frac{a_n}{n} = a_1 + \frac{S_a(x)}{x} + \int_1^x \frac{S_a(t)}{t^2} dt = \int_1^x \frac{S_a(t)}{t^2} dt + O(1).$$

Let us define $y = y(x) \geq 0$ nondecreasing so that:

1. $\lim_{x \rightarrow \infty} y = \infty$, and
2. $\log y = o(\log x)$.

For instance, $y(x) = \exp((\log x)^{1/2})$. Writing $S_a(x) = \alpha x + E(x)$, we have

$$\begin{aligned} \sum_{n \leq x} \frac{a_n}{n} &= \int_1^{y(x)} \frac{S_a(t)}{t^2} dt + \int_{y(x)}^x \frac{S_a(t)}{t^2} dt + O(1) \\ &= \int_{y(x)}^x \frac{\alpha t + E(t)}{t^2} dt + O(\log y(x)) \\ &= [\alpha \log t]_{y(x)}^x + \int_{y(x)}^x \frac{E(t)}{t^2} dt + o(\log x). \end{aligned}$$

Finally, $\int_{y(x)}^x \frac{|E(t)|}{t^2} dt \leq \left(\sup_{y(x) \leq t \leq x} \frac{|E(t)|}{t} \right) \int_{y(x)}^x \frac{dt}{t} = o(\log x)$. □

Example 29.4. Suppose that $\sum_{p \leq x} \log p \sim \alpha x$ with $\alpha > 0$. Then

$$\sum_{p \leq x} \frac{\log p}{p} \sim \alpha \log x.$$

30 Mertens' theorems

Theorem 30.1 (Mertens).

1.

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1).$$

2. There is a constant M such that

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O\left(\frac{1}{\log x}\right).$$

Proof. (1) Using $\log n = \sum_{d|n} \Lambda(d)$, we have

$$\sum_{n \leq x} \log n = \sum_{n \leq x} \sum_{d|n} \Lambda(d) = \sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right] = x \sum_{n \leq x} \frac{\Lambda(n)}{n} + O\left(\sum_{n \leq x} \Lambda(n)\right).$$

Using Chebyshev's theorem, we get

$$\sum_{p \leq x} \frac{\log p}{p} = \sum_{n \leq x} \frac{\Lambda(n)}{n} + O(1) = \log x + O(1).$$

(2) Let $S(x) = \sum_{p \leq x} \frac{\log p}{p}$. Partial summation gives

$$\sum_{p \leq x} \frac{1}{p} = \frac{1}{2} + \left[\frac{S(t)}{\log t} \right]_2^x + \int_2^x \frac{S(t)}{t(\log t)^2} dt = 1 + \int_2^x \frac{S(t)}{t(\log t)^2} dt + O\left(\frac{1}{\log x}\right).$$

Let us write $\sum_{p \leq x} \frac{\log p}{p} = \log x + R(x)$ so that $R(x) = O(1)$. Then

$$\int_2^x \frac{S(t)}{t(\log t)^2} dt = \int_2^x \frac{1}{t \log t} dt + \int_2^x \frac{R(t)}{t(\log t)^2} dt = [\log |\log t|]_2^x + O\left(\frac{1}{\log t}\right). \square$$

Corollary 30.2. *If the limit $\lim_{x \rightarrow \infty} \pi(x)/(x/\log x) = \alpha$ exists, then $\alpha = 1$.*

Proof. If the limit exists, then $\sum_{p \leq x} \vartheta(x) \sim \alpha x$. By Chebyshev's theorem, $\alpha > 0$. Then $\sum_{p \leq x} \frac{\log p}{p} \sim \alpha \log x$ so $\alpha = 1$ by above. $\square$

Remark 30.3. The above corollary was first proved by Chebyshev by a more elaborate method, before Mertens' theorems were available.

31 Bertrand's postulate

Theorem 31.1 (Chebyshev). *For all n sufficiently large, there exists a prime p satisfying between $n \leq p \leq 2n$.*

Proof. We follow the proof by S. Ramanujan. Let $a_0 \geq a_1 \geq \dots$ be a nondecreasing sequence with $\lim_{n \rightarrow \infty} a_n = 0$. Then we have

$$a_0 - a_1 = \sum_{n=0}^{\infty} (-1)^n a_n \leq a_0 - a_1 + a_2.$$

Using $\log n = \sum_{d|n} \Lambda(d)$, we get

$$T(x) := \sum_{n \leq x} \log n = \sum_{n \leq x} \sum_{d|n} \Lambda(d) = \sum_{e \leq x} \psi\left(\frac{x}{e}\right).$$

We have

$$\sum_{n \leq x} (-1)^{n-1} \psi\left(\frac{x}{n}\right) = T(x) - 2T\left(\frac{x}{2}\right) = (\log 2)x + O(\log x).$$

It follows that

$$\psi(x) - \psi\left(\frac{x}{2}\right) \leq (\log 2)x + O(\log x) \leq \psi(x) - \psi\left(\frac{x}{2}\right) + \psi\left(\frac{x}{3}\right).$$

From the first inequality, we get $\psi(x) \leq 2(\log 2)x + O((\log x)^2)$. Using this to bound $\psi(x/3)$ from above and using the second inequality, we get

$$\begin{aligned} \psi(x) - \psi\left(\frac{x}{2}\right) &\geq (\log 2)x + O(\log x) - \left(2(\log 2)\frac{x}{3} + O((\log(x/3))^2)\right) \\ &= \frac{\log 2}{3}x + O((\log x)^2). \end{aligned}$$

Since $\vartheta(x) = \psi(x) + O(\sqrt{x}(\log x)^2)$ we get

$$\vartheta(x) - \vartheta\left(\frac{x}{2}\right) = \frac{\log 2}{3}x + O(\sqrt{x}(\log x)^2).$$

□